

BCS-DBT Code Structure

Breast Cancer Detection System: DBT fine-tuning and 2D transfer learning. Repository snapshot 2026-10-04.

1. Overview

The code classifies **lesion presence** (abnormal = benign lesion + malignant; normal = no lesion) in 3D digital breast tomosynthesis (DBT), then transfers the trained model to 2D mammography datasets. One model class, `DimensionAgnostic3DSwin`, serves both stages: a 2D image is fed as a one-slice volume ($Z=1$). The project goal is reproducibility, so splits are deterministic and patient-grouped, seeds are fixed, and train sets are not resampled each epoch.

Pipeline

Step	What happens	Code
0. Cache	Decode each multiframe DICOM once into uint8 [Z,384,384] .npz on the SSD (/data/cache).	<code>fine_tuning/prewarm_cache.py</code>
1. (optional) SSL	MoCo v2 self-supervised pretraining of the Swin-B backbone on DBT slices.	<code>fine_tuning/pretrain_ssl.py</code>
2. DBT fine-tune	Supervised training on BCS-DBT + DBT-2026 combined; val selects the checkpoint, test reported once.	<code>fine_tuning_combined/fine-tune_combined.py</code>
3. 2D transfer	Load the DBT checkpoint and fine-tune on each 2D dataset (EMBED, VinDr, CMMD, INbreast, ...).	<code>transfer_learning/<dataset>/run.py</code>
4. Experiments	Multi-seed matrices, ablations, and AUROC summaries for the paper.	<code>experiments/*.py</code>

2. Directory layout

```
BreastCancerDetectionSystem/  
  CLAUDE.md  Dockerfile  docker-compose.yml  requirements.txt  main.py  
  common/  
    fine_tuning/  
    fine_tuning_combined/  
    transfer_learning/  
      embed/  vindr/  cmmd/  inbreast/  cbis/  mias/  busi/  rsna/  cddcesm/  
    experiments/  
    scripts/  
    docs/  logs/  results/  papers/  EMBED/tables/  
    checkpoints/  checkpoints_b16/  model weights (do not move)
```

3. Module reference

common/ (shared by both stages)

File	Lines	Role / main definitions
<code>model.py</code>	177	DimensionAgnostic3DSwin : timm Swin backbone (Swin-B 384 default, <code>in_chans=k</code> for 2.5D) run per slice in chunks, MIL max-pool over slices, WSOL saliency head (1x1 conv). Options: <code>ssl_weights</code> , <code>freeze_backbone</code> , <code>head_type</code> (mlp or CosineHead). Padded slices masked with <code>-inf</code> .
<code>preprocessing.py</code>	87	MedicalImagePreprocessor : DICOM read + VOI LUT, breast crop, CLAHE, aspect-preserving pad/resize to 384, box mask rendering.

fine_tuning/ (stage 1, BCS-DBT)

File	Lines	Role / main definitions
data_preprocessing.py	508	BCSDBTDataset (SSD cache, 2.5D neighbor slices, augmentation), collate_dbt_volumes (variable-Z padding + depth mask), samplers: UndersampleNormalSampler, BalancedBatchSampler, RandomRatioUndersampler; build_dbt_data_loader.
fine-tune.py	285	train_model: focal loss + gated Dice (WSOL), 3-group LLRD, bf16, early stopping, saliency_to_box.
pretrain_ssl.py	243	MoCo v2 SSL: DBTSliceSSL (cross-view / adjacent-slice positives), Encoder, MoCo; saves backbone only (ssl_swinb_backbone.pth).
prewarm_cache.py	70	Offline multi-worker cache builder per split.

fine_tuning_combined/ (stage 1, combined: current main)

File	Lines	Role / main definitions
combined.py	85	LesionBCSDBT (lesion label; Actionable dropped), build_combined_dataset (ConcatDataset per split), RatioUndersampler (normal:abnormal 9:1).
dataset_dbt2026.py	227	DBT2026Dataset : Segmed DBT-2026 loader, same output format as BCS-DBT; deterministic crc32 stratified study split; separate cache dir.
fine-tune_combined.py	181	train_combined: focal loss, LLRD + linear warmup + grad-clip 0.5, seeded workers/generator, val for selection, test held out. Writes checkpoints/combined_*.pth.
train_scratch.py	16	Same run from random init (ablation vs ImageNet init).

transfer_learning/ (stage 2, 2D)

File	Lines	Role / main definitions
mammo2d_base.py	152	Mammo2DDataset base: single-frame DICOM decode, SSD cache, emits [k,1,H,W] (Z=1) so the 3D model is reused; deterministic_bucket split; collate_mammo2d. Subclasses implement only _build_records().
transfer_common.py	352	run_transfer : dataset-agnostic train/eval loop; optional strict load of DBT checkpoint (else ImageNet baseline); focal loss, LLRD groups, Ratio/Balanced undersamplers, KD loss.
run_kd.py	37	Cross-dimension knowledge distillation: 3D DBT teacher to 2D student.
<dataset>/dataset.py	42-88	One subclass per dataset: EMBED (+ natural-ratio variant), VinDr, CMMD, INbreast, CBIS, MIAS, BUSI (ultrasound), RSNA, CDD-CESM. Patient-grouped splits.
<dataset>/run.py	13-14	Entry point: calls run_transfer with the dataset class and DBT checkpoint.
embed/ablation.py, llrd_ablation.py	15, 32	EMBED partial-freeze and LLRD ablations.

experiments/ (paper runs)

File	Lines	Role / main definitions
multiseed_matrix.py	125	methods x datasets x balancing x seeds; val-only selection, test once, saves predictions for DeLong; resumable; summarize mode.
malignancy_run / _matrix / _balanced_run.py	37-52	Malignancy track on malignancy-labelled datasets.
scratch_transfer.py, scratch_balanced_lesion.py	31-33	Transfer from the scratch-init DBT checkpoint.
cosine_m1_run.py	55	Cosine head + weighted CE ablation.
balanced_run.py, embed_nat_matrix.py	24-32	Balanced arm; EMBED natural-ratio matrix.
compare_m0_m1.py, organize_auc.py	50, 80	Bootstrap comparison ImageNet-TL vs DBT-TL; AUROC summary writer.

Root and scripts/

File	Lines	Role / main definitions
main.py	238	Early prototype (per-slice DICOM loader, malignant label). Superseded by fine_tuning/.
Dockerfile, docker-compose.yml	38, 36	A6000 container bc_detection_system_a6000; data mounted read-only.
scripts/smoke_test_25d.py	57	Forward-pass check for 2.5D inputs.
scripts/embed_finish.sh	12	EMBED download finishing helper.

4. Shared data contract

Every dataset returns (volume [k,Z,H,W], mask [1,Z,H,W], label). The collate function pads Z and returns {volumes, depth_mask, masks, labels}; the model returns per-slice logits [B,Z] (padding = -inf) and saliency maps [B,Z,H,W]. Scan score = max over slices. For 2D data Z=1, so the max is the identity and the same weights load strictly.

5. How to run

```
sg docker -c 'docker compose up -d'
sg docker -c 'docker exec -w /workspace -e PYTHONPATH=/workspace bc_detection_system_a6000 \
python fine_tuning_combined/fine-tune_combined.py'
sg docker -c 'docker exec -w /workspace -e PYTHONPATH=/workspace bc_detection_system_a6000 \
python transfer_learning/embed/run.py'
sg docker -c 'docker exec -w /workspace -e PYTHONPATH=/workspace bc_detection_system_a6000 \
python experiments/multiseed_matrix.py 42'
```

Long jobs: launch with docker exec -d and log to /workspace (not the exFAT SSD).